

Differential Equations Lesson 13

Question 1:

The general solution for the differential equation $y'' + 4y' + 3y = 0$ is $y(x) = C_1 \cdot e^{-x} + C_2 \cdot e^{-3x}$, where C_1 and C_2 are arbitrary constants.

Taylor series expansion for e^{-x} is $e^{-x} = 1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \frac{(-x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!}$

Taylor series expansion for e^{-3x} is $e^{-3x} = 1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$

The power series solution for the differential equation $y'' + 4y' + 3y = 0$, centered at $x_0 = 0$, is

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 0$ an ordinary point or a singular point?

b) Let $u_n = \frac{(-1)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-1)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

c) Let $u_n = \frac{(-3)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

d) The radius of convergence of the power series solution

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] \text{ is } \underline{\hspace{2cm}} ? \underline{\hspace{2cm}}.$$

e) The interval of convergence of the power series solution

$$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] \text{ is } \underline{\hspace{2cm}} ? \underline{\hspace{2cm}}.$$

Note: $\frac{n!}{(n+1)!} = \frac{n \cdot (n-1)(n-2)\dots 3 \cdot 2 \cdot 1}{(n+1)n \cdot (n-1)(n-2)\dots 3 \cdot 2 \cdot 1} = \frac{1}{n+1}$

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1} \cdot x^{n+1}}{(n+1)!}}{\frac{(-1)^n \cdot x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \cdot \frac{x^{n+1}}{x^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \cdot x \right| \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| \cdot \lim_{n \rightarrow \infty} |x| = 0 \cdot |x| = 0\end{aligned}$$

Hence, $\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^n}{n!}$ converges for any real number x .

Meaning radius of convergence is $(-\infty, \infty)$

Note: $\frac{n!}{(n+1)!} = \frac{n \cdot (n-1)(n-2)\dots 3 \cdot 2 \cdot 1}{(n+1)n \cdot (n-1)(n-2)\dots 3 \cdot 2 \cdot 1} = \frac{1}{n+1}$

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}}{\frac{(-3)^n \cdot x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \cdot \frac{3^{n+1} \cdot x^{n+1}}{3^n \cdot x^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \cdot 3x \right| \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| \cdot \lim_{n \rightarrow \infty} |3x| = 0 \cdot |3x| = 0\end{aligned}$$

Hence, $\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$ converges for any real number x .

Meaning radius of convergence is $(-\infty, \infty)$

Question 2:

The general solution for the differential equation $y'' + 6y' + 9y = 0$ is $y(x) = C_1 \cdot e^{-3x} + C_2 x \cdot e^{-3x}$,
where C_1 and C_2 are arbitrary constants.

Taylor series expansion for e^{-3x} is $e^{-3x} = 1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$

Taylor series expansion for $x \cdot e^{-3x}$ is $x \cdot e^{-3x} = x \left[1 + (-3x) + \frac{(-3x)^2}{2!} + \frac{(-3x)^3}{3!} + \frac{(-3x)^4}{4!} + \dots \right] = \sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!}$

The power series solution for the differential equation $y'' + 6y' + 9y = 0$, centered at $x_0 = 0$, is

$y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$, where a_0 and a_1 are arbitrary constants.

a) Is the point $x_0 = 0$ an ordinary point or a singular point?

b) Let $u_n = \frac{(-3)^n \cdot x^n}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

c) Let $u_n = \frac{(-3)^n \cdot x^{n+1}}{n!}$ and $u_{n+1} = \frac{(-3)^{n+1} \cdot x^{n+2}}{(n+1)!}$. $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \underline{\hspace{2cm}} ?$

d) The radius of convergence of the power series solution $y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$ is $\underline{\hspace{2cm}} ?$

e) The interval of convergence of the power series solution $y(x) = a_0 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!} \right] + a_1 \cdot \left[\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!} \right]$ is $\underline{\hspace{2cm}} ?$

Note: $\frac{n!}{(n+1)!} = \frac{n \cdot (n-1)(n-2) \dots 3 \cdot 2 \cdot 1}{(n+1)n \cdot (n-1)(n-2) \dots 3 \cdot 2 \cdot 1} = \frac{1}{n+1}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(-3)^{n+1} \cdot x^{n+1}}{(n+1)!}}{\frac{(-3)^n \cdot x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \cdot \frac{3^{n+1} \cdot x^{n+1}}{3^n \cdot x^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \cdot 3x \right| \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| \cdot \lim_{n \rightarrow \infty} |3x| = 0 \cdot |3x| = 0 \end{aligned}$$

Hence, $\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^n}{n!}$ converges for any real number x .

Meaning radius of convergence is $(-\infty, \infty)$

Note: $\frac{n!}{(n+1)!} = \frac{n \cdot (n-1)(n-2) \dots 3 \cdot 2 \cdot 1}{(n+1)n \cdot (n-1)(n-2) \dots 3 \cdot 2 \cdot 1} = \frac{1}{n+1}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(-3)^{n+1} \cdot x^{n+2}}{(n+1)!}}{\frac{(-3)^n \cdot x^{n+1}}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \cdot \frac{3^{n+1} \cdot x^{n+2}}{x^{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \cdot 3x \right| \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| \cdot \lim_{n \rightarrow \infty} |3x| = 0 \cdot |3x| = 0 \end{aligned}$$

Hence, $\sum_{n=0}^{\infty} \frac{(-3)^n \cdot x^{n+1}}{n!}$ converges for any real number x .

Meaning radius of convergence is $(-\infty, \infty)$

Question 3:

The power series solution for the differential equation $y'' + 6y' + 5y = 0$, centered at $x_0 = 1$, is

$$y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n, \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 1$ an ordinary point or a singular point?

b) Does the differential equation $y'' + 6y' + 5y = 0$ have any singular points?

c) The radius of convergence of the power series solution $y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n$ is _____?

d) The interval of convergence of the power series solution $y(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5a_0 + a_1) - (-5)^n (a_0 + a_1)}{4 \cdot n!} (x-1)^n$ is _____?

The differential equation $y'' + 6y' + 5y = 0$ is of the form $y'' + P(x)y' + Q(x) \cdot y = 0$.

Hence, $P(x) = 6$ and $Q(x) = 5$.

A singular point occurs where either $P(x)$ or $Q(x)$ fails to be analytic.

Also, if $P(x_0)$ is undefined or $Q(x_0)$ is undefined then x_0 is a singular point.

Since $P(x)$ and $Q(x)$ are analytic everywhere, $y'' + 6y' + 5y = 0$

does not have singular points.

Question 4:

The power series solution for the differential equation $y'' + 16y = 0$, centered at $x_0 = 2$, is

$$y(x) = a_0 \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 16^n}{(2n)!} (x-2)^{2n} \right] + a_1 \left[\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 16^n}{(2n+1)!} (x-2)^{2n+1} \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

- a) Is the point $x_0 = 2$ an ordinary point or a singular point?
- b) Does the differential equation $y'' + 16y = 0$ does not have any singular points?
- c) The radius of convergence of the power series solution $y(x)$ is ____?
- d) The interval of convergence of the power series solution $y(x)$ is ____?

Question 5:

Radius of convergence and interval of convergence of power series solution:

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point.

The power series solution for the differential equation $y'' + 4y' + \frac{1}{x-1} \cdot y = 0$, centered at $x_0 = 3$, is

$$y(x) = a_0 \left[1 - \frac{1}{4}(x-3)^2 + \frac{3}{8}(x-3)^3 - \frac{3}{8}(x-3)^4 + \frac{93}{320}(x-3)^4 \dots \right] + a_1 \left[(x-3) - 2(x-3)^2 + \frac{31}{12}(x-3)^3 - \frac{119}{48}(x-3)^4 \dots \right],$$

where a_0 and a_1 are arbitrary constants.

- a) Is the point $x_0 = 3$ an ordinary point or a singular point?
- b) Is the point $x_0 = 1$ a regular point or a singular point?
- c) The radius of convergence of the power series solution $y(x)$ is ____?
- d) The interval of convergence of the power series solution $y(x)$ is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

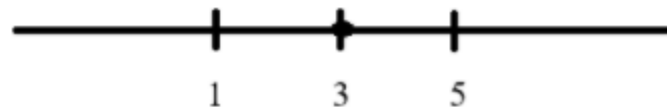
between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

The radius of convergence is the distance from $x = 3$ to $x = 1$, which is 2.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

$$(3 - 2, 3 + 2) = (1, 5)$$



Question 6:

Radius of convergence and interval of convergence of power series solution:

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point.

The power series solution for the differential equation $y'' + \frac{4}{x^2 - 4} y' + y = 0$, centered at $x_0 = 1$, is

$$y(x) = a_0 \left[1 - \frac{1}{2}(x-1)^2 - \frac{2}{9}(x-1)^3 - \frac{23}{216}(x-1)^4 - \dots \right] + a_1 \left[(x-1) + \frac{2}{3}(x-1)^2 + \frac{5}{18}(x-1)^3 + \frac{2}{9}(x-1)^4 + \dots \right],$$

where a_0 and a_1 are arbitrary constants.

- a) Is the point $x_0 = 1$ an ordinary point or a singular point?
- b) Is the point $x_0 = 2$ an ordinary point or a singular point?
- c) Is the point $x_0 = -2$ an ordinary point or a singular point?
- d) The radius of convergence of the power series solution $y(x)$ is ____?
- e) The interval of convergence of the power series solution $y(x)$ is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

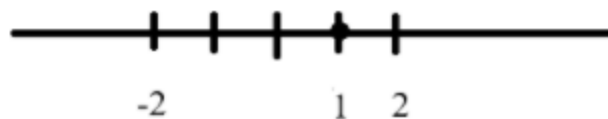
between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

The radius of convergence is the distance from $x = 1$ to $x = 2$, which is 1.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

$$(1 - 1, 1 + 1) = (0, 2)$$



Question 7:

The power series solution for the differential equation $y'' + \frac{1}{x^2 + 1}y' + y = 0$, centered at $x_0 = 4$, is

$$y(x) = a_0 \left[1 - \frac{1}{2}(x-4)^2 + \frac{1}{102}(x-4)^3 + \frac{2}{51}(x-4)^4 - \dots \right] \\ + a_1 \left[(x-4) - \frac{1}{34}(x-4)^2 - \frac{140}{867}(x-4)^3 + \frac{9}{2312}(x-4)^4 \dots \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 4$ an ordinary point or a singular point?

b) The singular points for the differential equation $y'' + \frac{1}{x^2 + 1}y' + y = 0$ are ____?

c) The radius of convergence of the power series solution $y(x)$ is ____?

Hint: The distance between two complex numbers, $z_1 = a_1 + b_1 \cdot i$ and $z_2 = a_2 + b_2 \cdot i$, is $d = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2}$.

The distance from $x = 4 = 4 + 0i$ to $x = i = 0 + 1i$ is $\sqrt{(0 - 4)^2 + (1 - 0)^2} = \sqrt{17}$

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

$$(4 - R, 4 + R) = (4 - \sqrt{17}, 4 + \sqrt{17}).$$

Question 8:

The power series solution for the differential equation $y'' + 4y' + \frac{1}{x^2 + 9}y = 0$, centered at $x_0 = 1$, is

$$y(x) = a_0 \left[1 - \frac{1}{20}(x-1)^2 + \frac{7}{100}(x-1)^3 - \frac{829}{12000}(x-1)^4 + \dots \right] \\ + a_1 \left[(x-1) - 2(x-1)^2 + \frac{53}{20}(x-1)^3 - \frac{1579}{600}(x-1)^4 + \dots \right], \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}$$

a) Is the point $x_0 = 1$ an ordinary point or a singular point?

b) The singular points for the differential equation $y'' + 4y' + \frac{1}{x^2 + 9}y = 0$ are ____?

c) The radius of convergence of the power series solution $y(x)$ is ____?

Hint: The distance from $x = 1 = 1 + 0i$ to $x = 3i = 0 + 3i$ is $\sqrt{(0-1)^2 + (3-0)^2} = \sqrt{10}$

d) The interval of convergence of the power series solution $y(x)$ is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Radius of convergence is $(1 - R, 1 + R) = (1 - \sqrt{10}, 1 + \sqrt{10})$.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 9:

Find the series solution of the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$, centered at the ordinary point $x = 0$.

Power series solution for the differential equation is assumed to have the form: $y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$

The objective is to find the values of $a_0, a_1, a_2, a_3, a_4, a_5, \dots$

First, we need to find $y'(x)$ and $y''(x)$.

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots$$

$$y''(x) = 2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots, +$$

Substituting $y(x)$, $y'(x)$, and $y''(x)$ into the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$:

$$[2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots] + 3x^2(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots) + x(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) = 0$$

$$[2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots] + (3a_1 x^2 + 3 \cdot 2a_2 x^3 + 3 \cdot 3a_3 x^4 + 3 \cdot 4a_4 x^5 + 3 \cdot 5a_5 x^6 + \dots) + (a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + a_4 x^5 + \dots) = 0 + 0x + 0x^2 + 0x^3 + 0x^4 + \dots$$

Comparing the coefficients of corresponding powers of x on both sides:

For terms with no x (or x^0): $2a_2 = 0 \Rightarrow a_2 = 0$

For terms with x^1 : $2 \cdot 3a_3 + a_0 = 0 \Rightarrow a_3 = -\frac{1}{6}a_0$

For terms with x^2 : $3 \cdot 4a_4 + 3 \cdot 1a_1 + a_1 = 0$

For terms with x^3 : $4 \cdot 5a_5 + 3 \cdot 2a_2 + a_2 = 0$

For terms with x^4 : $5 \cdot 6a_6 + 3 \cdot 3a_3 + a_3 = 0$

For terms with x^5 : $6 \cdot 7a_7 + 3 \cdot 4a_4 + a_4 = 0$

For terms with x^n : $(n+1) \cdot (n+2) \cdot a_{n+2} + 3 \cdot (n-1)a_{n-1} + a_{n-1} = 0$

Solving for a_{n+2} : $a_{n+2} = -\frac{3(n-1)a_{n-1} + a_{n-1}}{(n+1)(n+2)} = -\frac{[3n-3+1]a_{n-1}}{(n+1)(n+2)} = -\frac{[3n-2]a_{n-1}}{(n+1)(n+2)}$

For $n=0$: $a_2 = -\frac{[3(0)-2]a_{-1}}{(0+1)(0+2)} = \frac{[3(0)-2] \cdot 0}{(0+1)(0+2)} = 0$

Note: a_{-1} is not part of the power series solution $y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$

Any coefficient with a negative index like a_{-1} is defined as zero.

For terms with x^5 :

exponent over x is 5

$$\begin{array}{rcl}
 6 \cdot 7a_7 & + & 3 \cdot 4a_4 + a_4 = 0 \\
 \downarrow & & \downarrow \quad \downarrow \quad \downarrow \\
 7 = 5 + 2 & & 4 = 5 - 1 \quad 4 = 5 - 1 \\
 7 = 5 + 2 & & 3 \text{ remains constant} \\
 6 = 5 + 1 & &
 \end{array}$$

Recurrence relation

$$\text{For } n = 1: a_3 = -\frac{[3(1) - 2]a_0}{(1+1)(1+2)} = -\frac{1}{6}a_0$$

$$\text{For } n = 2: a_4 = -\frac{[3(2) - 2]a_1}{(2+1)(2+2)} = -\frac{1}{3}a_1$$

$$\text{For } n = 3: a_5 = -\frac{[3(3) - 2]a_2}{(3+1)(3+2)} = -\frac{7}{20}a_2 = -\frac{7}{20} \cdot 0 = 0$$

$$\text{For } n = 4: a_6 = -\frac{[3(4) - 2]a_3}{(4+1)(4+2)} = -\frac{1}{3}a_3 = -\frac{1}{3}\left(-\frac{1}{6}a_0\right) = \frac{1}{18}a_0$$

$$\text{For } n = 5: a_7 = -\frac{[3(5) - 2]a_4}{(5+1)(5+2)} = -\frac{13}{42}a_4 = -\frac{13}{42}\left(-\frac{1}{3}a_1\right) = \frac{13}{126}a_1$$

$$\text{For } n = 6: a_8 = -\frac{[3(6) - 2]a_5}{(6+1)(6+2)} = -\frac{16}{56}a_5 = -\frac{16}{56} \cdot 0 = 0$$

a) Power Series Solution for the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$ is

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + a_7 x^7 + a_8 x^8 + \dots$$

$$y(x) = \underline{\hspace{2cm}} ?$$

Note: Power series solution will be in terms of a_0 and a_1 ;
where a_0 and a_1 are arbitrary constants.

- b) Does the differential equation $y'' + 3x^2 \cdot y' + x \cdot y = 0$ have singular points?
- c) The radius of convergence of the power series solution $y(x)$ is ____?
- d) The interval of convergence of the power series solution $y(x)$ is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 10:

Find the series solution of the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$, centered at the ordinary point $x = 0$.

Power series solution for the differential equation is assumed to have the form: $y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$

The objective is to find the values $a_0, a_1, a_2, a_3, a_4, a_5, \dots$

First, we need to find $y'(x)$ and $y''(x)$.

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots$$

$$y''(x) = 2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots$$

Substituting $y(x)$, $y'(x)$, and $y''(x)$ into the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$:

$$(1+x^2)(2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots) + 4x(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots) + 4x^2(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) = 0$$

$$\Rightarrow (2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots) + x^2(2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots) + 4x(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots) + 4x^2(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) = 0$$

$$\Rightarrow (2a_2 + 2 \cdot 3a_3 x + 3 \cdot 4a_4 x^2 + 4 \cdot 5a_5 x^3 + \dots) + (1 \cdot 2a_2 x^2 + 2 \cdot 3a_3 x^3 + 3 \cdot 4a_4 x^4 + 4 \cdot 5a_5 x^5 + \dots) + (4 \cdot 1a_1 x + 4 \cdot 2a_2 x^2 + 4 \cdot 3a_3 x^3 + 4 \cdot 4a_4 x^4 + 4 \cdot 5a_5 x^5 + \dots) + (4a_0 x^2 + 4a_1 x^3 + 4a_2 x^4 + 4a_3 x^5 + 4a_4 x^6 + \dots) = 0 + 0x + 0x^2 + 0x^3 + 0x^4 + \dots$$

Comparing the coefficients of corresponding powers of x on both sides:

$$\text{For terms with no } x \text{ (or } x^0 \text{): } 2a_2 = 0 \quad \Rightarrow \quad a_2 = 0$$

$$\text{For terms with } x^1: \quad 2 \cdot 3a_3 + 4 \cdot 1a_1 = 0 \quad \Rightarrow \quad a_3 = \frac{-4}{6}a_1$$

$$\text{For terms with } x^2: \quad 3 \cdot 4a_4 + 1 \cdot 2a_2 + 4 \cdot 2a_2 + 4 \cdot a_0 = 0$$

$$\text{For terms with } x^3: \quad 4 \cdot 5a_5 + 2 \cdot 3a_3 + 4 \cdot 3a_3 + 4 \cdot a_1 = 0$$

$$\text{For terms with } x^4: \quad 5 \cdot 6a_6 + 3 \cdot 4a_4 + 4 \cdot 4a_4 + 4 \cdot a_2 = 0$$

$$\text{For terms with } x^5: \quad 6 \cdot 7a_7 + 4 \cdot 5a_5 + 4 \cdot 5a_5 + 4 \cdot a_3 = 0$$

$$\text{For terms with } x^n: \quad (n+1) \cdot (n+2) \cdot a_{n+2} + (n-1) \cdot n \cdot a_n + 4 \cdot n \cdot a_n + 4 \cdot a_{n-2} = 0$$

$$\text{Solving for } a_{n+2}: \quad a_{n+2} = -\frac{(n^2 - n + 4n)a_n + 4a_{n-2}}{(n+1)(n+2)} = -\frac{(n^2 + 3n)a_n + 4a_{n-2}}{(n+1)(n+2)}$$

$$\text{For } n=0: \quad a_2 = -\frac{(0^2 + 3 \cdot 0)a_0 + 4a_{-2}}{(0+1)(0+2)} = 0$$

Note: a_{-2} is not part of the power series solution $y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$.

Any coefficient with a negative index like a_{-2} is defined as zero.

$$\text{For } n=1: \quad a_3 = -\frac{(1^2 + 3 \cdot 1)a_1 + 4a_{-1}}{(1+1)(1+2)} = -\frac{4a_1 + 4 \cdot 0}{(1+1)(1+2)} = -\frac{4a_1}{6} = -\frac{2}{3}a_1$$

Note: a_{-1} is not part of the power series solution $y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$.

Any coefficient with a negative index like a_{-1} is defined as zero.

$$\text{For } n=2: a_4 = -\frac{(2^2 + 3 \cdot 2)a_2 + 4a_0}{(2+1)(2+2)} = -\frac{(2^2 + 3 \cdot 2) \cdot 0 + 4a_0}{12} = -\frac{1}{3}a_0$$

$$\text{For } n=3: a_5 = -\frac{(3^2 + 3 \cdot 3)a_3 + 4a_1}{(3+1)(3+2)} = -\frac{18a_3 + 4a_1}{20} = -\frac{18\left(-\frac{2}{3}a_1\right) + 4a_1}{20} = \frac{2}{5}a_1$$

$$\text{For } n=4: a_6 = -\frac{(4^2 + 3 \cdot 4)a_4 + 4a_2}{(4+1)(4+2)} = -\frac{28a_4 + 4a_2}{30} = -\frac{28\left(-\frac{1}{3}a_0\right) + 4 \cdot 0}{30} = \frac{14}{45}a_0$$

$$\text{For } n=5: a_7 = -\frac{(5^2 + 3 \cdot 5)a_5 + 4a_3}{(5+1)(5+2)} = -\frac{40a_5 + 4a_3}{42} = -\frac{40 \cdot \left(\frac{2}{5}a_1\right) + 4 \cdot \left(-\frac{2}{3}a_1\right)}{42} = \frac{20}{63}a_1$$

$$\text{For } n=6: a_8 = -\frac{(6^2 + 3 \cdot 6)a_6 + 4a_4}{(6+1)(6+2)} = -\frac{54a_6 + 4a_4}{56} = -\frac{54\left(\frac{14}{45}a_0\right) + 4\left(-\frac{1}{3}a_0\right)}{56} = -\frac{29}{105}a_0$$

a) Power Series Solution for the differential equation $(1+x^2)y'' + 4xy' + 4x^2y = 0$ is

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$y(x) = \underline{\hspace{2cm}} ?$$

Note: Power series solution will be in terms of a_0 and a_1 ;
where a_0 and a_1 are arbitrary constants.

b) The singular points of the differential equation $(1+x^2)y'' + 4x \cdot y' + 4x^2 \cdot y = 0$ are ____?

$$\text{Hint: } (1+x^2)y'' + 4x \cdot y' + 4x^2 \cdot y = 0 \Rightarrow y'' + \frac{4x}{(1+x^2)} \cdot y' + \frac{4x^2}{(1+x^2)} \cdot y = 0$$

$$P(x) = \frac{4x}{(1+x^2)} \quad \text{and} \quad Q(x) = \frac{4x^2}{(1+x^2)}$$

Singular points are where $P(x)$ and $Q(x)$ are undefined; so set $1+x^2 = 0$ and solve for x .

c) The radius of convergence of the power series solution $y(x)$ is ____?

d) The interval of convergence of the power series solution $y(x)$ is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.